CSA0251

C Programming for recursion

HOMEWORK - 3

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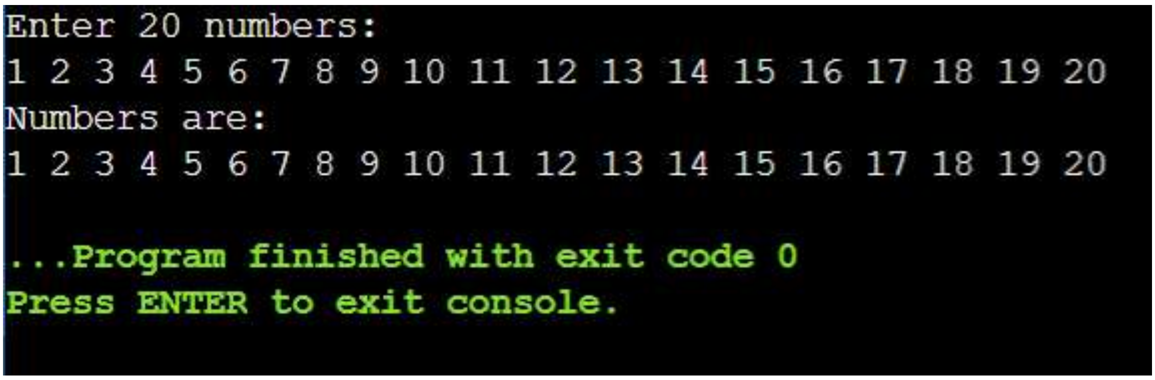
1. Write a C program to Read and display 20 numbers using array and run time input.

PROGRAM :

```
#include <stdio.h>
int main() {
   int a[20], i;
   printf("Enter 20 numbers:\n");
   for(i=0;i<20;i++)
      scanf("%d",&a[i]);
    printf("Numbers are:\n");
   for(i=0;i<20;i++)
      printf("%d ",a[i]);
   return 0;
}
```

OUTPUT:



```
Enter 20 numbers:
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20
Numbers are:
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20

...Program finished with exit code 0
Press ENTER to exit console.
```

2. Write a C program to Get 20 numbers using array initialization and display these 20 numbers using array.

PROGRAM:

```c
#include <stdio.h>

int main() {

  int a[20] = {

    1,2,3,4,5,

    6,7,8,9,10,

    11,12,13,14,15,

    16,17,18,19,20

  };

  int i;

  printf("Array elements are:\n");

  for(i = 0; i < 20; i++) {

    printf("%d ", a[i]);

  }

  return 0;

}
```

OUTPUT :

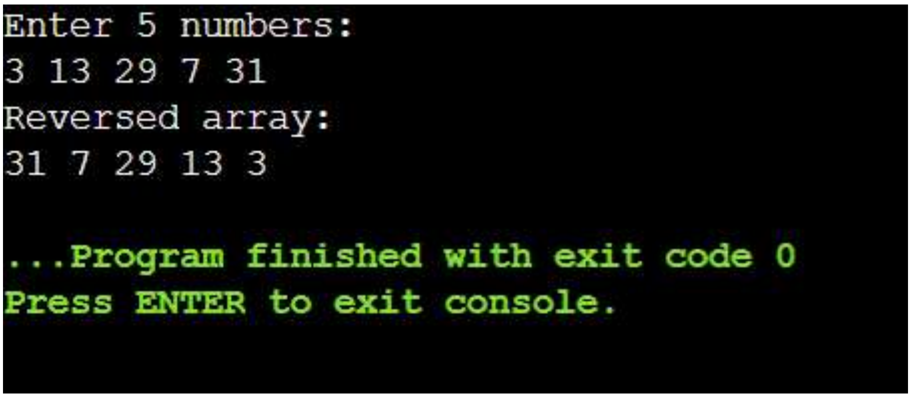
```
Array elements are:
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20

...Program finished with exit code 0
Press ENTER to exit console.
```

3. Write a program to reverse the elements of an array.

PROGRAM:

```
#include <stdio.h>
int main() {
   int a[5], i;
   printf("Enter 5 numbers:\n");
   for(i=0;i<5;i++)
      scanf("%d",&a[i]);
   printf("Reversed array:\n");
   for(i=4;i>=0;i--)
      printf("%d ",a[i]);
   return 0;
}
```

OUTPUT:



```
Enter 5 numbers:
3 13 29 7 31
Reversed array:
31 7 29 13 3

...Program finished with exit code 0
Press ENTER to exit console.
```

4. Write a C program to find the average of 25 numbers using array.

PROGRAM:

```
#include <stdio.h>
int main() {
   int a[25], i, sum=0
   printf("Enter 25 numbers:\n");
   for(i=0;i<25;i++) {
      scanf("%d",&a[i]);
      sum += a[i];
   }
   printf("Average = %.2f", (float)sum/25);
   return 0;
}
```

OUTPUT:

```
Enter 25 numbers:
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25
Average = 13.00

...Program finished with exit code 0
Press ENTER to exit console.
```

5. Write a C program to find smallest array element in C.

PROGRAM :

```
#include <stdio.h>

int main() {

  int a[100], n, i, min;

  printf("Enter number of elements:\n");

  scanf("%d", &n);

  printf("Enter elements:\n");

  for(i = 0; i < n; i++) {

    scanf("%d", &a[i]);

  }

  min = a[0];

  for(i = 1; i < n; i++) {

    if(a[i] < min) {

      min = a[i];

    }

  }

  printf("Smallest element = %d", min);

  return 0;

}
```

OUTPUT:

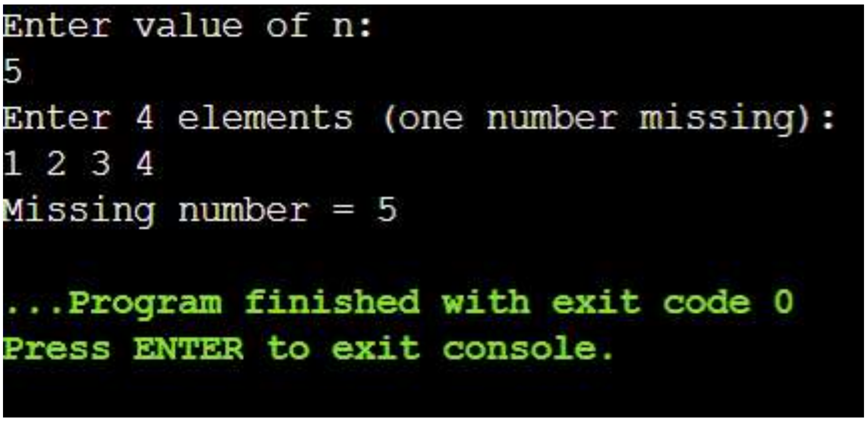
```
Enter number of elements:
5
Enter elements:
3 5 7 9 11
Smallest element = 3

...Program finished with exit code 0
Press ENTER to exit console.
```

6. Write a C Program to find Missing elements

PROGRAM :

```
#include <stdio.h>

int main() {

  int n, i, sum = 0, total;

  printf("Enter value of n:\n");

  scanf("%d", &n);

  int a[n-1];

  printf("Enter %d elements (one number missing):\n", n-1);

  for(i = 0; i < n-1; i++) {

    scanf("%d", &a[i]);

    sum += a[i];

  }

  total = n * (n + 1) / 2;

  printf("Missing number = %d", total - sum);

  return 0;

}
```

OUTPUT:

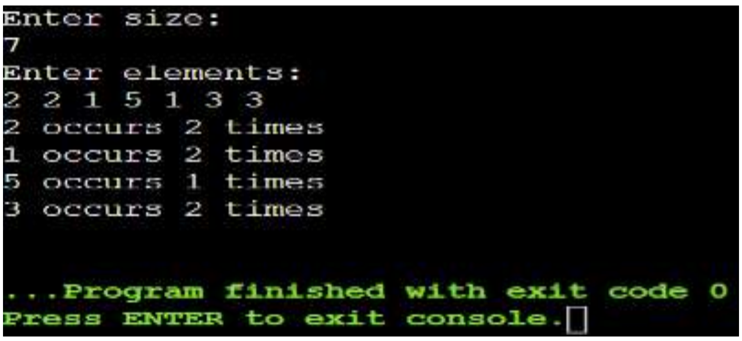


7. Write a C program to count the frequency of each element in an array.

PROGRAM :

```
#include <stdio.h>

int main() {

  int a[100], n, i, j, count, visited[100]={0};

  scanf("%d",&n);

  printf("Enter elements:\n");

  for(i=0;i<n;i++)

    scanf("%d",&a[i]);

  for(i=0;i<n;i++) {

    if(visited[i]) continue;

    count = 1;

    for(j=i+1;j<n;j++) {

      if(a[i]==a[j]) {

        count++;

        visited[j]=1;

      }

    printf("%d occurs %d times\n", a[i], count);

  }

  return 0;

}
```

OUTPUT:



```
Enter size:
7
Enter elements:
2 2 1 5 1 3 3
2 occurs 2 times
1 occurs 2 times
5 occurs 1 times
3 occurs 2 times


...Program finished with exit code 0
Press ENTER to exit console.
```

8. Write a C program to find majority elements in an array..

PROGRAM :

```c
#include <stdio.h>

int main() {

  int a[100], n, i, j, count;

  printf("Enter size:\n");

  scanf("%d",&n);

  printf("Enter elements:\n");

  for(i=0;i<n;i++)

    scanf("%d",&a[i]);

  for(i=0;i<n;i++) {

    count=0;

    for(j=0;j<n;j++)

      if(a[i]==a[j]) count++;

    if(count > n/2) {

      printf("Majority element = %d", a[i]);

      return 0;

    }

  }

  printf("No majority element");

  return 0;

}
```

OUTPUT:

```
Enter size:
5
Enter elements:
2 2 2 4 5
Majority element = 2

...Program finished with exit code 0
Press ENTER to exit console.
```

9. Write a C Program to delete an element in the array.

PROGRAM :

```
#include <stdio.h>
int main() {
  int a[100], n, i, pos;
  printf("Enter size:\n");
  scanf("%d",&n);
  printf("Enter elements:\n");
  for(i=0;i<n;i++)
    scanf("%d",&a[i]);
  printf("Enter position to delete:\n");
  scanf("%d",&pos);
  for(i=pos-1;i<n-1;i++)
    a[i]=a[i+1];
  printf("Updated array:\n");
  for(i=0;i<n-1;i++)
    printf("%d ",a[i]);
  return 0;
}
```

OUTPUT:

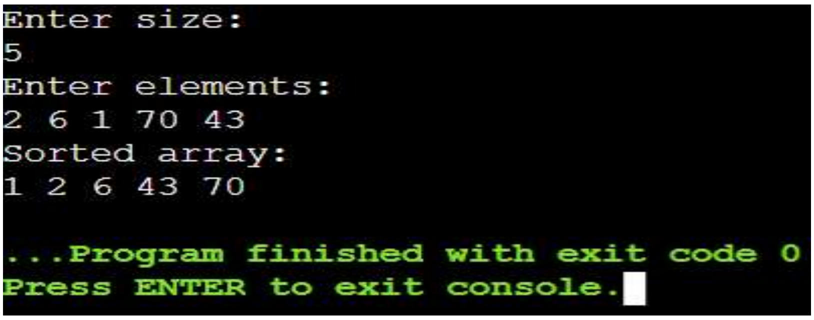
```
Enter size:
5
Enter elements:
2 4  6 7 8
Enter position to delete:
4
Updated array:
2 4 6 8

...Program finished with exit code 0
Press ENTER to exit console.
```

10. Write a C Program to sort array in ascending order.

PROGRAM :

```
#include <stdio.h>
int main() {
  int a[100], n, i, j, temp;
  printf("Enter size:\n");
  scanf("%d",&n);
  printf("Enter elements:\n");
  for(i=0;i<n;i++)
    scanf("%d",&a[i])
  for(i=0;i<n;i++)
    for(j=i+1;j<n;j++)
      if(a[i]>a[j]) {
        temp=a[i];
        a[i]=a[j];
        a[j]=temp;
      }
  printf("Sorted array:\n");
  for(i=0;i<n;i++)
    printf("%d ",a[i]);
  return 0;
}
```

OUTPUT:



```
Enter size:
5
Enter elements:
2 6 1 70 43
Sorted array:
1 2 6 43 70

...Program finished with exit code 0
Press ENTER to exit console.
```

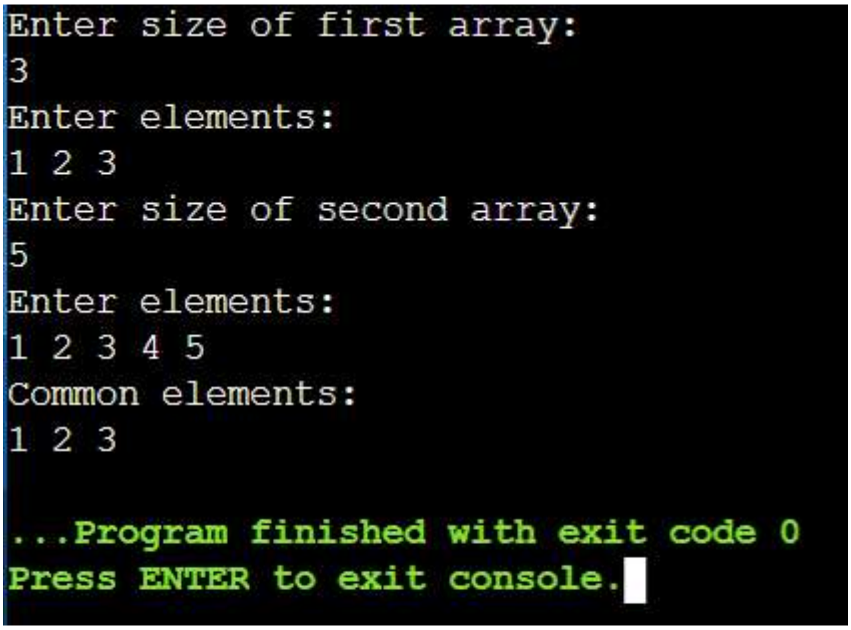
11.Write a program to find the common element(s) in two arrays.

PROGRAM :

```
#include <stdio.h>
int main() {
   int a[100], b[100];
   int n1, n2, i, j, found = 0;
   printf("Enter size of first array:\n");
   scanf("%d", &n1);
   printf("Enter elements of first array:\n");
   for(i = 0; i < n1; i++) {
      scanf("%d", &a[i]);
   }
   printf("Enter size of second array:\n");
   scanf("%d", &n2);
   printf("Enter elements of second array:\n");
   for(i = 0; i < n2; i++) {
      scanf("%d", &b[i]);
   }
   printf("Common elements are:\n");
   for(i = 0; i < n1; i++) {
      for(j = 0; j < n2; j++) {
         if(a[i] == b[j]) {
            printf("%d ", a[i]);
            found = 1;
            break;
         }
      }
```

```
    }
    if(found == 0) {
        printf("No common elements");
    }
    return 0;
}
```

OUTPUT:



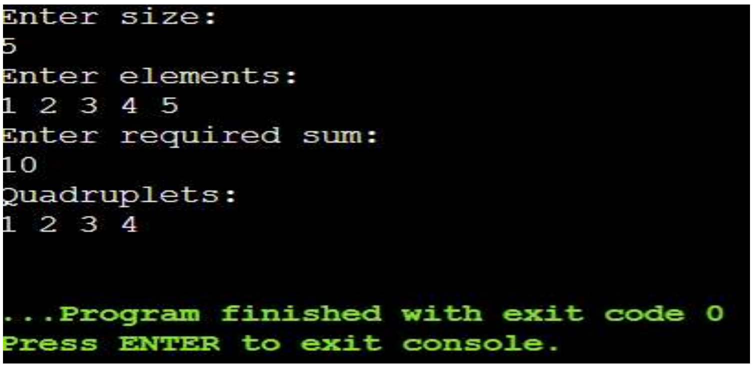
```
Enter size of first array:
3
Enter elements:
1 2 3
Enter size of second array:
5
Enter elements:
1 2 3 4 5
Common elements:
1 2 3

...Program finished with exit code 0
Press ENTER to exit console.
```

12.Write a C program to find all unique quadruplets in an array of sum up to a given value.

PROGRAM :

```
#include <stdio.h>
int main() {
  int a[50], n, i,j,k,l, sum
  printf("Enter size:\n");
  scanf("%d",&n)
  printf("Enter elements:\n");
  for(i=0;i<n;i++) scanf("%d",&a[i]);
  printf("Enter required sum:\n");
  scanf("%d",&sum)
  printf("Quadruplets:\n");
  for(i=0;i<n;i++)
  for(j=i+1;j<n;j++)
  for(k=j+1;k<n;k++)
  for(l=k+1;l<n;l++)
    if(a[i]+a[j]+a[k]+a[l]==sum)
      printf("%d %d %d %d\n",a[i],a[j],a[k],a[l]);
  return 0;
}
```

OUTPUT:



```
Enter size:
5
Enter elements:
1 2 3 4 5
Enter required sum:
10
Quadruplets:
1 2 3 4


...Program finished with exit code 0
Press ENTER to exit console.
```

13.Write a C program to find the triplets in an array whose sum equals a given value.

PROGRAM :

```c
#include <stdio.h>

int main() {

  int a[50], n, i,j,k, sum;

  printf("Enter size:\n");

  scanf("%d",&n);

  printf("Enter elements:\n");

  for(i=0;i<n;i++) scanf("%d",&a[i])

  printf("Enter required sum:\n");

  scanf("%d",&sum)

  printf("Triplets:\n");

  for(i=0;i<n;i++)

  for(j=i+1;j<n;j++)

  for(k=j+1;k<n;k++)

    if(a[i]+a[j]+a[k]==sum)

      printf("%d %d %d\n",a[i],a[j],a[k]);

  return 0;

}
```

OUTPUT:

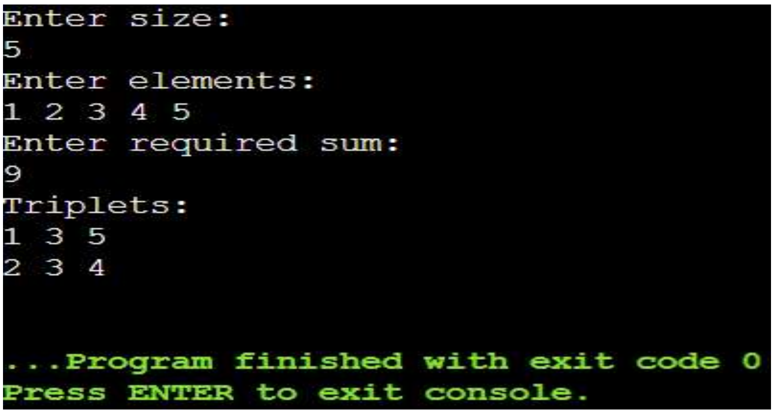
```
Enter size:
5
Enter elements:
1 2 3 4 5
Enter required sum:
9
Triplets:
1 3 5
2 3 4


...Program finished with exit code 0
Press ENTER to exit console.
```

14.Write a C program to find the triplets in an array whose sum equals a given value.

PROGRAM :

```
#include <stdio.h>
int main() {
  int a[50], n, i,j,k, sum;
  printf("Enter size:\n");
  scanf("%d",&n);
  printf("Enter elements:\n");
  for(i=0;i<n;i++) scanf("%d",&a[i])
  printf("Enter required sum:\n");
  scanf("%d",&sum)
  printf("Triplets:\n");
  for(i=0;i<n;i++)
  for(j=i+1;j<n;j++)
  for(k=j+1;k<n;k++)
    if(a[i]+a[j]+a[k]==sum)
      printf("%d %d %d\n",a[i],a[j],a[k]);
  return 0;
}
```

OUTPUT:



15. Write a C program to find the second largest element in an array.

PROGRAM :

```c
#include <stdio.h>

int main() {

  int a[100], n, i, max, second;

  printf("Enter size:\n");

  scanf("%d",&n);

  printf("Enter elements:\n");

  for(i=0;i<n;i++) scanf("%d",&a[i]);

  max = second = -9999;

  for(i=0;i<n;i++) {

    if(a[i] > max) {

      second = max;

      max = a[i];

    } else if(a[i] > second && a[i] != max) {

      second = a[i];

    }

  }

  printf("Second largest = %d", second);

  return 0;

}
```

OUTPUT:

```
Enter size:
5
Enter elements:
10 20 30 40 50
Second largest = 40

...Program finished with exit code 0
Press ENTER to exit console.
```

16.Write a C program to count all the pairs with a given difference.

PROGRAM :

```
#include <stdio.h>
int main() {
  int a[50], n, i, j, diff;
  printf("Enter size:\n");
  scanf("%d",&n);
  printf("Enter elements:\n");
  for(i=0;i<n;i++) scanf("%d",&a[i]);
  printf("Enter difference:\n");
  scanf("%d",&diff);
  printf("Pairs:\n");
  for(i=0;i<n;i++) {
    for(j=i+1;j<n;j++) {
      if((a[i]-a[j]==diff) || (a[j]-a[i]==diff))
        printf("(%d,%d)\n",a[i],a[j]);
    }
  }
  return 0;
}
```

OUTPUT:

```
Enter size:
5
Enter elements:
1 2 3 4 5
Enter difference:
2
Pairs:
(1,3)
(2,4)
(3,5)


...Program finished with exit code 0
Press ENTER to exit console.
```

17.write a program to find the union of two arrays.

PROGRAM :

```
#include <stdio.h>

int main() {

  int a[50], b[50], n, m, i, j, found;

  scanf("%d",&n);

  for(i=0;i<n;i++) scanf("%d",&a[i]);

  scanf("%d",&m);

  for(i=0;i<m;i++) scanf("%d",&b[i]);

  printf("Union:\n");

  for(i=0;i<n;i++)

    printf("%d ",a[i]);

  for(i=0;i<m;i++) {

    found = 0;

    for(j=0;j<n;j++) {

      if(b[i] == a[j]) {

        found = 1;

        break;

      } if(found == 0)

      printf("%d ",b[i]);

  } return 0;

}
```

OUTPUT:

```
Enter size of first array:
5
Enter elements:
1 2 3 4 5
Enter size of second array:
5
Enter elements:
3 4 5 6 7
Union:
1 2 3 4 5 6 7

...Program finished with exit code 0
Press ENTER to exit console.
```

18. write a program to move all zeroes in an array to end while maintaining the order of elements.

PROGRAM :

```
#include <stdio.h
int main() {
  int a[50], n, i, j = 0;
  printf("Enter size:\n");
  scanf("%d", &n);
  printf("Enter elements:\n");
  for(i = 0; i < n; i++) {
    scanf("%d", &a[i]);
  } for(i = 0; i < n; i++) {
    if(a[i] != 0) {
      a[j] = a[i];
      j++;
    }
  } while(j < n) {
    a[j] = 0;
    j++;
  } printf("Result:\n");
  for(i = 0; i < n; i++) {
    printf("%d ", a[i]);
  } return 0;
}
```

OUTPUT:

```
Enter size:
9
Enter elements:
0 1 9 8 4 0 2 7 0
After moving zeros:
1 9 8 4 2 7 0 0 0

...Program finished with exit code 0
Press ENTER to exit console.
```

19. Write a C program to count all the pairs with a given difference.

PROGRAM :

```
#include <stdio.h>

int main() {

  int a[50], n, i, j, diff, count = 0;

  printf("Enter size:\n");

  scanf("%d", &n)

  printf("Enter elements:\n");

  for(i = 0; i < n; i++) {

    scanf("%d", &a[i]);

  }

  printf("Enter difference:\n");

  scanf("%d", &diff);

  for(i = 0; i < n; i++) {

    for(j = i + 1; j < n; j++) {

      if(a[i] - a[j] == diff || a[j] - a[i] == diff) {

        count++;

      }

    }

  } printf("Number of pairs = %d", count);

  return 0;

}
```

OUTPUT:

```
Enter size:
5
Enter elements:
1 2 3 4 5
Enter difference:
2
Number of pairs = 3

...Program finished with exit code 0
Press ENTER to exit console.
```

20. Write a C Program to find the Kth smallest and Kth largest element in a given array.

PROGRAM :

```c
#include <stdio.h>

int main() {

  int a[50], n, i, j, temp, k;

  scanf("%d", &n);

  printf("Enter elements:\n");

  for(i = 0; i < n; i++) {

    scanf("%d", &a[i]);

  }

  printf("Enter k:\n");

  scanf("%d", &k);

  for(i = 0; i < n; i++) {

    for(j = i + 1; j < n; j++) {

      if(a[i] > a[j]) {

        temp = a[i];

        a[i] = a[j];

        a[j] = temp;

      }

  } printf("Kth smallest = %d\n", a[k-1]);

  printf("Kth largest = %d", a[n-k]);

  return 0;

}
```

OUTPUT: